

# BOLUN XU

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## POSITIONS

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**Columbia University**, NY, US

*January 2020 - Present*

Assistant Professor

Earth and Environmental Engineering & Electrical Engineering (affiliated)

**Massachusetts Institute of Technology**, MA, US

*July 2018 - December 2019*

Postdoctoral Associate

MIT Energy Initiative and Lab for Info. & Decision Systems

## EDUCATION

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**University of Washington**, Seattle, US

*September 2014 - June 2018*

PhD in Electrical Engineering

**Swiss Federal Institute of Technology**, Zurich, CH

*September 2011 - January 2014*

MS in Electrical Engineering

**Shanghai Jiaotong University**, Shanghai, China

*September 2007 - August 2011*

BS in Electrical and Computer Engineering (Dual degree with **University of Michigan**, Ann Arbor)

## JOURNAL PUBLICATIONS

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1. Zhiyuan Fan and **Bolun Xu**, “Design optimization and global impact assessment of solar-thermal direct air carbon capture,” *Environmental Science and Technology*, 2026
2. Ning Qi, Yousuf Baker, and **Bolun Xu**, “Online convex optimization for coordinated long-term and short-term isolated microgrid dispatch,” *Applied Energy*, 2026
3. Zhiyuan Fan, E. Dentzer, J. Glynn, D. S. Goldberg, J. Friedmann, and **Bolun Xu**, “Enhancing profit and co2 mitigation: Commercial direct air capture design and operation with power market volatility,” *Engineering*, 2026
4. Liudong Chen and **Bolun Xu**, “A prudent framework for understanding risk-awareness in demand response,” *European Journal of Operational Research*, 2025
5. Xueyuan Cui, Y. Wang, and **Bolun Xu**, “Dimension-reduced optimization of multi-zone thermostatically controlled loads,” *IEEE Transactions on Smart Grid*, vol. 16, no. 6, pp. 4685–4697, 2025
6. Ning Qi and **Bolun Xu**, “Locational energy storage bid bounds for facilitating social welfare convergence,” *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 3, no. 4, pp. 486–497, 2025
7. Ming Yi, Saud Alghumayjan, and **Bolun Xu**, “Perturbed decision-focused learning for modeling strategic energy storage,” *IEEE Transactions on Smart Grid*, 2025
8. Ning Qi, K. Huang, Zhiyuan Fan, and **Bolun Xu**, “Long-term energy management for microgrid with hybrid hydrogen-battery energy storage: A prediction-free coordinated optimization framework,” *Applied Energy*, vol. 377, p. 124485, 2025
9. **Bolun Xu** and B. F. Hobbs, “On truthful pricing of battery energy storage resources in electricity spot markets,” in *Oxford Energy Forum*, vol. 140, pp. 34–38, 2024

10. Saud Alghumayjan, Jiajun Han, Ningkun Zheng, Ming Yi, and **Bolun Xu**, “Energy storage arbitrage in two-settlement markets: A transformer-based approach,” *Electric Power Systems Research*, vol. 235, p. 110755, 2024
11. T. A. Hansen, E. J. Wilson, J. P. Fitts, M. Jansen, P. Beiter, B. Steffen, **Bolun Xu**, J. Guillet, M. Münster, and L. Kitzing, “Five grand challenges of offshore wind financing in the united states,” *Energy Research & Social Science*, vol. 107, p. 103329, 2024
12. N. Nazir, V. Bruscas, Y. Dvorkin, J. Lidicker, S. Polunsky, P. Pugliese, W. Tonis, **Bolun Xu**, and M. Kintner-Meyer, “Transit fleet electrification barriers, resolutions and costs,” *Findings*, 2023
13. Y. Bian, Ningkun Zheng, Y. Zheng, **Bolun Xu**, and Y. Shi, “Predicting strategic energy storage behaviors,” *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1608–1619, 2024
14. Y. Shi and **Bolun Xu**, “Demand-side price-responsive flexibility and baseline estimation through end-to-end learning,” *IET Renewable Power Generation*, vol. 18, no. 3, pp. 361–371, 2024
15. Joshua Jaworski, Ningkun Zheng, M. Preindl, and **Bolun Xu**, “Vehicle-to-grid fleet service provision considering nonlinear battery behaviors,” *IEEE Transactions on Transportation Electrification*, vol. 10, no. 2, pp. 2945–2955, 2024
16. X. Qin, **Bolun Xu**, I. Lestas, Y. Guo, and H. Sun, “The role of electricity market design for energy storage in cost-efficient decarbonization,” *Joule*, vol. 7, no. 6, pp. 1227–1240, 2023
17. Yousuf Baker, Ningkun Zheng, and **Bolun Xu**, “Transferable energy storage bidder,” *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 4117–4126, 2023
18. Ningkun Zheng, X. Qin, D. Wu, G. Murtaugh, and **Bolun Xu**, “Energy storage state-of-charge market model,” *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 1, no. 1, pp. 11–22, 2023
19. **Bolun Xu**, “The role of modeling battery degradation in bulk power system optimizations,” *MRS Energy & Sustainability*, pp. 1–14, 2022
20. Umar Salman, S. Belaish, Z. Ji, D. Huang, Ningkun Zheng, and **Bolun Xu**, “Comparing the economic value of lithium-ion battery technologies in the nine wholesale electricity markets in north america,” *iEnergy*, vol. 1, no. 3, pp. 363–373, 2022
21. Ningkun Zheng, Joshua Jaworski, and **Bolun Xu**, “Arbitraging variable efficiency energy storage using analytical stochastic dynamic programming,” *IEEE Transactions on Power Systems*, vol. 37, no. 6, pp. 4785–4795, 2022
22. **Bolun Xu**, “Dynamic valuation of battery lifetime,” *IEEE Transactions on Power Systems*, vol. 37, no. 3, pp. 2177–2186, 2021
23. I. Mathews, **Bolun Xu**, W. He, V. Barreto, T. Buonassisi, and I. M. Peters, “Technoeconomic model of second-life batteries for utility-scale solar considering calendar and cycle aging,” *Applied Energy*, vol. 269, p. 115127, 2020
24. Y. Liu, **Bolun Xu**, A. Botterud, N. Zhang, and C. Kang, “Bounding regression errors in data-driven power grid steady-state models,” *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1023–1033, 2020
25. **Bolun Xu**, Y. Shi, D. S. Kirschen, and B. Zhang, “Optimal battery participation in frequency regulation markets,” *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6715–6725, 2018
26. **Bolun Xu**, J. Zhao, T. Zheng, E. Litvinov, and D. S. Kirschen, “Factoring the cycle aging cost of batteries participating in electricity markets,” *IEEE Transactions on Power Systems*, vol. 33, no. 2, pp. 2248–2259, 2017

27. Y. Dvorkin, R. Fernandez-Blanco, Y. Wang, **Bolun Xu**, D. S. Kirschen, H. Pandžić, J.-P. Watson, and C. A. Silva-Monroy, “Co-planning of investments in transmission and merchant energy storage,” *IEEE Transactions on Power Systems*, vol. 33, no. 1, pp. 245–256, 2017
28. Y. Wang, Y. Dvorkin, R. Fernandez-Blanco, **Bolun Xu**, T. Qiu, and D. S. Kirschen, “Look-ahead bidding strategy for energy storage,” *IEEE Transactions on Sustainable Energy*, vol. 8, no. 3, pp. 1106–1117, 2017
29. Y. Shi, **Bolun Xu**, D. Wang, and B. Zhang, “Using battery storage for peak shaving and frequency regulation: Joint optimization for superlinear gains,” *IEEE Transactions on Power Systems*, vol. 33, no. 3, pp. 2882–2894, 2017
30. **Bolun Xu**, Y. Wang, Y. Dvorkin, R. Fernández-Blanco, C. A. Silva-Monroy, J.-P. Watson, and D. S. Kirschen, “Scalable planning for energy storage in energy and reserve markets,” *IEEE Transactions on Power systems*, vol. 32, no. 6, pp. 4515–4527, 2017
31. **Bolun Xu**, A. Oudalov, A. Ulbig, G. Andersson, and D. S. Kirschen, “Modeling of lithium-ion battery degradation for cell life assessment,” *IEEE Transactions on Smart Grid*, vol. 9, no. 2, pp. 1131–1140, 2016
32. R. Fernández-Blanco, Y. Dvorkin, **Bolun Xu**, Y. Wang, and D. S. Kirschen, “Optimal energy storage siting and sizing: A wecc case study,” *IEEE Transactions on Sustainable Energy*, vol. 8, no. 2, pp. 733–743, 2016
33. T. Qiu, **Bolun Xu**, Y. Wang, Y. Dvorkin, and D. S. Kirschen, “Stochastic multistage coplanning of transmission expansion and energy storage,” *IEEE Transactions on Power Systems*, vol. 32, no. 1, pp. 643–651, 2016

## CONFERENCE PAPERS

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1. Emily Logan, Ning Qi, and **Bolun Xu**, “Nonparametric kernel regression for coordinated energy storage peak shaving with stacked services,” in *2026 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2026
2. Saud Alghumayjan, Ming Yi, and **Bolun Xu**, “A few-shot LLM framework for extreme day classification in electricity markets,” in *2026 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2026
3. Liudong Chen, F. Zhao, and **Bolun Xu**, “The limitations of proxy bus models in coordinated transaction scheduling,” in *Proceedings of the 17th ACM International Conference on Future and Sustainable Energy Systems*, pp. 1–10, 2026
4. Yousuf Baker, Ning Qi, Ming Yi, and **Bolun Xu**, “Bellman-consistent reduced-dimension representation of energy storage value functions for accelerated computation,” in *Proceedings of the 17th ACM International Conference on Future and Sustainable Energy Systems*, pp. 1–10, 2026
5. Y. Wu, Ming Yi, **Bolun Xu**, and J. Anderson, “Online energy storage arbitrage under imperfect predictions: A conformal risk-aware approach,” in *Proceedings of the 17th ACM International Conference on Future and Sustainable Energy Systems*, pp. 1–10, 2026
6. Liudong Chen, H. Kim, A. Elmachoub, and **Bolun Xu**, “Fair aggregation in virtual power plants,” in *2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026
7. Y. Du and **Bolun Xu**, “From semi-scheduled to firm: A comparative review and roadmap for mitigating vre uncertainty in the australian nem,” in *2026 IEEE GreenTech*, 2026
8. Elizabeth Cohn, Ning Qi, U. Lall, and **Bolun Xu**, “A lagrangian-informed long-term dispatch policy for coupled hydropower and photovoltaic systems,” in *2025 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2025

9. Neal Ma, Ningkun Zheng, Ning Qi, and **Bolun Xu**, “Comparative withholding behavior analysis of historical energy storage bids in california,” in *2025 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2025
10. Y. Wu, **Bolun Xu**, and J. Anderson, “Energy storage arbitrage under price uncertainty: Market risks and opportunities,” in *2025 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2025
11. Saud Alghumayjan, Ming Yi, and **Bolun Xu**, “Conformal uncertainty quantification of electricity price predictions for risk-averse storage arbitrage,” in *2025 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2025
12. Y. Wu, **Bolun Xu**, and J. Anderson, “Market power and withholding behavior of energy storage units,” *2024 IEEE 63rd Annual Conference on Decision and Control (CDC)*, 2024
13. Zhiyi Zhou, Ningkun Zheng, R. Zhang, and **Bolun Xu**, “Energy storage market power withholding bounds in real-time markets,” in *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems*, pp. 215–225, 2024
14. Liudong Chen, X. Zhu, **Bolun Xu**, and F. Ding, “Demand side flexibility envelope quantification under data scarcity,” in *2024 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*, pp. 1–5, IEEE, 2024
15. Liudong Chen and **Bolun Xu**, “Saturation effects in equitable demand response tariff design,” in *2023 North American Power Symposium (NAPS)*, pp. 1–5, IEEE, 2023
16. Ningkun Zheng, X. Liu, **Bolun Xu**, and Y. Shi, “Energy storage price arbitrage via opportunity value function prediction,” *IEEE Power and Energy Society General Meeting*, 2023
17. Ningkun Zheng and **Bolun Xu**, “Impact of bidding and dispatch models over energy storage utilization in bulk power systems,” in *IREP Symposium on Bulk Power System Dynamics and Control*, vol. 2022, 2022
18. Y. Bian, Ningkun Zheng, Y. Zheng, **Bolun Xu**, and Y. Shi, “Demand response model identification and behavior forecast with optnet: a gradient-based approach,” in *Proceedings of the Thirteenth ACM International Conference on Future Energy Systems*, pp. 418–429, 2022
19. **Bolun Xu**, “Experiences on incorporating market experiments into energy system education,” in *2021 IEEE Integrated STEM Education Conference (ISEC)*, pp. 127–131, IEEE, 2021
20. Wenting Ma and **Bolun Xu**, “A data-driven nonlinear recharge controller for energy storage in frequency regulation,” in *2021 IEEE Power & Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2021
21. **Bolun Xu**, M. Korpås, and A. Botterud, “Operational valuation of energy storage under multi-stage price uncertainties,” in *2020 59th IEEE Conference on Decision and Control (CDC)*, pp. 55–60, IEEE, 2020
22. **Bolun Xu**, M. Korpås, A. Botterud, and F. OSullivan, “A lagrangian policy for optimal energy storage control,” in *2020 American Control Conference (ACC)*, pp. 224–230, IEEE, 2020
23. Y. Wang, Y. Dvorkin, R. Fernández-Blanco, **Bolun Xu**, and D. S. Kirschen, “Impact of local transmission congestion on energy storage arbitrage opportunities,” in *2017 IEEE Power & Energy Society General Meeting*, pp. 1–5, IEEE, 2017
24. **Bolun Xu**, Y. Shi, D. S. Kirschen, and B. Zhang, “Optimal regulation response of batteries under cycle aging mechanisms,” in *2017 IEEE 56th Annual Conference on Decision and Control (CDC)*, pp. 751–756, IEEE, 2017

25. Y. Shi, **Bolun Xu**, Y. Tan, and B. Zhang, “A convex cycle-based degradation model for battery energy storage planning and operation,” in *2018 Annual American Control Conference (ACC)*, pp. 4590–4596, IEEE, 2018
26. Y. Shi, **Bolun Xu**, B. Zhang, and D. Wang, “Leveraging energy storage to optimize data center electricity cost in emerging power markets,” in *Proceedings of the Seventh International Conference on Future Energy Systems*, pp. 1–13, 2016
27. **Bolun Xu**, Y. Dvorkin, D. S. Kirschen, C. A. Silva-Monroy, and J.-P. Watson, “A comparison of policies on the participation of storage in us frequency regulation markets,” in *2016 IEEE Power and Energy Society General Meeting (PESGM)*, pp. 1–5, IEEE, 2016
28. **Bolun Xu**, A. Oudalov, J. Poland, A. Ulbig, and G. Andersson, “Bess control strategies for participating in grid frequency regulation,” *IFAC Proceedings Volumes*, vol. 47, no. 3, pp. 4024–4029, 2014
29. **Bolun Xu**, A. Ulbig, and G. Andersson, “Impacts of dynamic line rating on power dispatch performance and grid integration of renewable energy sources,” in *IEEE PES ISGT Europe 2013*, pp. 1–5, IEEE, 2013

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## MANUSCRIPTS IN PREPARATION

1. Ruonan Pi, Zhiyuan Fan, and **Bolun Xu**, “Electric arc furnaces scheduling under electricity price volatility with reinforcement learning,” *arXiv preprint arXiv:2512.09293 (submitted to IEEE Transactions on Sustainable Energy)*, 2025
2. Liudong Chen, J. Sethuraman, and **Bolun Xu**, “Gaming on coincident peak shaving: Equilibrium and strategic behavior,” *arXiv preprint arXiv:2501.02792 (submitted to submitted to Manufacturing & Service Operations Management (MSOM))*, 2025
3. J. Anunrojwong, S. R. Balseiro, O. Besbes, and **Bolun Xu**, “Battery operations in electricity markets: Strategic behavior and distortions,” *arXiv preprint arXiv:2406.18685 (submitted to submitted to Manufacturing & Service Operations Management (MSOM))*, 2024
4. X. Qin, I. Lestas, and **Bolun Xu**, “Economic capacity withholding bounds of competitive energy storage bidders,” *arXiv preprint arXiv:2403.05705 (submitted to Manufacturing & Service Operations Management (MSOM))*, 2024
5. Zhiyuan Fan, T. Lin, and **Bolun Xu**, “Voluntary renewable programs: Optimal pricing and revenue allocation,” *arXiv preprint arXiv:2603.17190 (submitted to Managemnt Science)*, 2026
6. Eliza Cohn, Ning Qi, U. Lall, and **Bolun Xu**, “Real-time coordination of cascaded hydroelectric generation under decision-dependent uncertainties,” *arXiv preprint arXiv:2603.17931 (submitted to IEEE Transactions on Sustainable Energy)*, 2026
7. Xueyuan Cui, Liudong Chen, Y. Wang, and **Bolun Xu**, “Online scheduling of thermostatically controlled loads: A stepwise reformulation approach,” *submitted to IEEE Transactions on Smart Grid*, 2026
8. Liudong Chen and **Bolun Xu**, “Fairness as an investment: Dynamic participation and long-run profit in virtual power plants,” *submitted to IEEE Transactions on Energy Markets, Policy, and Regulation*, 2026

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## INVITED SEMINARS

- Princeton University, NJ, May 2026
- University of Hong Kong, China, April 2026
- University of California Berkeley, CA, November 2025

- Stanford University, CA, November 2025
- University of Michigan, Ann Arbor, MI, October 2025
- City College of New York, NY, October 2025
- Massachusetts Institute of Technology, MA, October 2025
- Washington State University, October 2025
- Electric Power Research Institute, CA, July 2025
- Lawrence Berkeley National Lab, CA, July 2025
- Shanghai Jiaotong University, China, July 2025
- Swiss Federal Institute of Technology Zurich (ETHZ), Switzerland, June 2025
- Swiss Federal Institute of Technology Lausanne (EPFL), Switzerland, June 2025
- Imperial College London, UK, March 2025
- University of Cambridge, UK, March 2025
- University of Texas, Dallas, October 2024
- University of Tennessee, Knoxville, October 2024
- Iowa State University, September 2024
- Tsinghua-Berkeley Shenzhen Institute, June 2024
- Cornell University, October 2023
- Columbia Business School, September 2023
- Tsinghua University, August 2023
- RWTH Aachen University, July 2023
- University of Texas Austin, April 2023
- University of Houston, October 2022
- Temple University, March 2022
- EPRI ISO/RTO Energy Storage Market Modeling Technical WG, December 2021
- Johns Hopkins University Energy Seminar, December 2020
- Stanford Smart Grid Seminar, November 2020

## **INVITED WORKSHOPS AND PANELS**

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- IEEE PES Transmission & Distribution Conference, Chicago, IL, 2026
- China-America Frontiers of Engineering Symposium, Hong Kong, China, 2026
- INFORMS Optimization Society Conference, Atlanta, GA, 2026
- INFORMS Annual Meeting, Atlanta, GA, 2025
- New York State Offshore Wind Academic Symposium, New York, NY, 2025
- IEEE PES General Meeting, Austin, TX, 2025
- Future Power Markets Forum, Washington, DC, 2025

- U.S. Department of Energy Sponsored Workshop on Unleashing the Power of Data and AI in Distribution Grid Operation and DER Management, New York, NY, 2024
- INFORMS Annual Meeting, Seattle, WA, 2024
- CircuBatt Chat Workshop, Montreal, Canada, 2024
- INFORMS Annual Meeting, Phoenix, AZ, 2023
- NREL Workshop on Autonomous Energy Systems, Golden, CO, 2023
- IEEE PES General Meeting, Orlando, FL, 2023
- US-European Joint NSF Workshop on Flexible Electric Grid and Critical Infrastructure for Resilient Society, Philadelphia, PA, 2023
- NSF Workshop on Carbon-neutral Electricity and Mobility, College Station, TX, 2023
- Transportation Research Board Annual Meeting, Washington, DC, 2023
- IEEE PES General Meeting, Denver, CO, 2022

## AWARDS

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- Outstanding Reviewer, *IEEE Transactions on Energy Markets, Regulation, and Policy*, 2024
- Early Career Best Paper Award, *Institute for Operations Research and the Management Sciences (INFORMS) Energy, Natural Resources, and the Environment (ENRE) Section*, 2023
- Outstanding Young Investigator Award, *Institute of Industrial & Systems Engineers (IISE) Energy Systems Division*, 2023
- CAREER Award, *National Science Foundation*, 2022
- Outstanding Reviewer, *IEEE Transactions on Power Systems*, 2020, 2022, 2023
- Outstanding Reviewer, *IEEE Transactions on Sustainable Energy*, 2018
- Scientific Achievement Award, *Clean Energy Institute, University of Washington*, 2018
- Graduate Fellowship, *Clean Energy Institute, University of Washington*, 2015
- Graduate Fellowship, *Grainger Foundation, University of Washington*, 2014
- Best Poster Award, *IEEE 4th European ISGT Conference at Copenhagen*, 2013
- Excellence Student Scholarship, *Shanghai Jiaotong University*, 2008

## EXTERNAL GRANTS

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1. Co-PI (Columbia Lead), Department of Energy, “Carrizo Pumped Storage Hydropower: Seasonal Storage for Low-Cost and Reliable Power”, \$14M (My budget: \$516,136), 2025-2028. Lead: New Mexico State University.
2. PI (single PI), Carbontech Development Initiative, “Concept Design and Assessment of Solar-thermal/Electric Hybrid Low-temperature Direct Air Carbon Capture”, \$80,000, 2025-2026.
3. Co-PI, Carbontech Development Initiative, “To Develop Membrane-free Electrolyzers & Systems for Conversion of Carbon Dioxide from Waste-to-Energy Plants into Sustainable Aviation Fuel”, \$250k (My budget: \$100k), 2025-2028. Lead: Columbia University.
4. Co-PI (Columbia Lead), NYSERDA, “Integrating Refrigerated Trailers and Thermal Storage into NY Power Grid”, \$981,931 (My budget: \$221k), 2025-2028. Lead: Energy One Solutions.

5. Co-PI, ARPA-E, “Low Temperature Membraneless Carbon Dioxide Electrolyzers for Low-cost Production of Ethanol under Dynamic Conditions”, \$4.76M (My budget: \$500k), 2024-2027. Lead: Columbia University.
6. PI (Single PI), Carbon Direct Inc., Gift grant for research in the area of carbon management, \$12,254, 2024.
7. Co-PI, National Science Foundation, “Collaborative Research: Material Simulation-driven Electrolyte Designs in Intermediate-temperature Na-K / S Batteries for Long-duration Energy Storage”, \$386,259 (My budget: \$50k), 2024-2027. Lead: Columbia University.
8. Co-PI (Columbia Lead), Department of Energy, “Western-based Analysis of Distributed Battery Storage System Benefits and Tradeoffs”, \$1,500k (My budget: \$375k), 2024-2027. Lead: Johns Hopkins University.
9. Co-PI (Columbia Lead), Department of Energy, “Smart Energy Storage Integration and Management Platform for Buildings (SESIMP-B)”, \$1,500k (My budget: \$300k), 2024-2027. Lead: Gridscape Solutions.
10. PI (single PI), Concord New Energy, “Machine learning for energy storage bidding in real-time wholesale energy markets”, \$650k, 2023-2028.
11. PI (single PI), National Science Foundation, “CAREER: Computation-efficient Algorithms for Grid-scale Energy Storage Control, Bidding, and Integration Analysis”, \$500,557, 2023-2028.
12. Co-PI (Columbia Lead), Department of Energy, “Advanced ISO Models for Storage and Hybrid Resources Operation”, \$300k (My budget: \$100k), 2023-2024. Lead institution: Lawrence Berkeley National Lab.
13. Co-PI (Columbia Lead), Department of Energy, “Assessing Energy Storage Bidding Models in CAISO Wholesale Electricity Markets”, \$80k (My budget: \$80k), 2022-2023. Lead: Pacific Northwest National Lab.

## INTERNAL GRANTS

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1. Co-PI, Columbia University, “Designing an Affordable and Just Residential Electricity Tariff System for All”, \$170k (My budget: \$85k), 2026-2028. Co-investigator: Adam Elmachetou.
2. PI (Single PI), Columbia University Office of the Provost Junior Faculty Grants, “Integrating Demographic Data to Equitable Energy Infrastructure Access and Electrification Analysis: A New York City Case Study”, \$30,000, 2024.
3. Co-PI, Columbia SEAS Interdisciplinary Research Seed (SIRS) Funding, “Intermediate-Temperature Na-K/S Batteries for Long-duration Energy Storage”, \$170k (My budget: \$85k), 2023-2025.
4. PI, Columbia Data Science Institute Seed Grant, “Positioning Energy Storage Technologies under Stochastic Climate Scenarios”, \$150k (My budget: \$150k), 2021-2023.

## INDUSTRY EXPERIENCE

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**Doosan Gridtech**, WA USA  
Power System Research Engineer Intern

*June 2017 - September 2017*

**ISO New England**, MA USA  
Research Intern Business Architecture & Technology Group

*June 2016 - August 2016*

**China Electric Power Research Institute**, Beijing China  
Research Intern in Distributed Energy Resource Group

*February 2014 - August 2014*

## **INDUSTRY ADVISOR ROLES**

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- Applied IQ LTD, board member, 2025-
- Shandong Kehui Power Automation Co Ltd, non-executive board member, 2023-
- Storlytics, Advisor, 2022-
- Sensai Analytics, Advisor, 2021-
- Recurrent Motors, Advisor, 2020-2021

## **EDUCATIONAL ACTIVITIES**

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- Faculty Advisor, Department of Energy Hydropower Collegiate Competition, 2023-
- Review committee and faculty advisor, Columbia University Summer Undergraduate Research Experience (SURE) Program, 2023-2025
- Faculty advisor, Columbia Summer@SEAS Program, 2022-
- Guest lecturer, Trevor Day School, 2025; Stuyvesant High School, 2020

## **DEPARTMENT SERVICES**

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- Chair of the undergraduate committee, Earth and Environmental Engineering, 2024-
- ABET representative, Earth and Environmental Engineering, 2024-2025
- Committee of Instruction (COI) representative, Earth and Environmental Engineering, 2022-2025
- Member of Diversity, Equity, and Inclusion (DEI) committee, Earth and Environmental Engineering, 2022-2025
- Chair of colloquium committee, Earth and Environmental Engineering, 2020-2022

## **PROFESSIONAL ENGAGEMENTS**

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- Associate Editor, IEEE Transactions on Power Systems, 2024-
- Technical program committee, ACM International Conference on Future and Sustainable Energy Systems (ACM e-Energy), 2023-
- Organizing committee, Columbia Forum: Smart Electric Energy and Transport, 2023-
- Officer, Working Group on Business Models for Energy Storage, IEEE Power & Energy Society Power System Operation, Planning, and Economics (PSOPE) committee, 2023-
- Member, Task Force on Data Analytics for Energy Storage, IEEE Power & Energy Society Subcommittee on Big Data & Analytics for Power Systems, 2022-
- Member, Working Group on Distribution System BTM DERs: Visibility, Analytics and Control, IEEE Power & Energy Society Distribution System Analysis Subcommittee, 2022-
- Guest Editor, IET Renewable Power Generation, Special Issue on: Computational Methods and Artificial Intelligence Applications in Low Carbon Energy Systems, 2022-2023
- Organizing committee, MIT A+B Applied Energy Symposium, 2020
- Review panelist for National Science Foundation (NSF), Department of Energy (DOE), and Climate Change AI

- Reviewer for Joule, IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid, IEEE Transactions on Sustainable Energy, IEEE IAS, Applied Energy, IET Generation, Transmission & Distribution, PES General Meeting, CDC, IFAC